

Systemic Therapy for Advanced Human Epidermal Growth Factor Receptor 2–Positive Breast Cancer: ASCO Guideline Update Q and A

Jennie R. Crews, MD¹; Jeffrey J. Kirshner, MD²; and Sarah Temin, MSPH³

ASCO recently updated the guideline for systemic therapy for patients with human epidermal growth factor receptor 2 (HER2)–positive advanced breast cancer, along with an updated guideline for the management of advanced human epidermal growth factor receptor 2 (HER2)–positive breast cancer and brain metastases.¹ On the basis of new evidence, recommendations for the optimal management in the second-line and third-line setting were updated. This companion article focuses on questions that clinicians might have as they implement these new recommendations for patients with HER2-positive advanced breast cancer who have progressed during or after first-line, second-line, or greater HER2-targeted therapy (Fig 1).

Question: What Is the Key Takeaway for Clinicians?

Jennie Crews, MD: Clinicians should offer trastuzumab deruxtecan as second-line therapy to patients with HER2-positive metastatic breast cancer and should monitor patients for interstitial lung disease and manage per guidelines.

Question: Why Did the Expert Panel Recommend Trastuzumab Deruxtecan Based on Progression-Free Survival?

Jeffrey Kirshner, MD: The drug was recommended based on the highly statistically significant and

clinically meaningful improvement in progression-free survival although there is not long enough follow-up to conclude that there is an improvement in overall survival.

Question: Will the Current Status of the Agent in the United States Potentially Affect Access to It?

Jeffrey Kirshner, MD: Although trastuzumab deruxtecan is not technically approved as “second-line” therapy for advanced HER2-positive disease, it is approved as third-line therapy, and given the impressive results of the DESTINY-03 study,^{2,3} it is likely to be approved by most insurance companies for these patients.

Question: What Is Needed for This Guideline to Become a Routine Part of Care?

Jennie Crews, MD: Given the significant improvement in progression-free survival, trend toward improved overall survival, and manageable side effects of trastuzumab deruxtecan in second-line metastatic HER2-positive breast cancer, this guideline should be adopted as routine care.

Author affiliations and support information (if applicable) appear at the end of this article.

Accepted on June 27, 2022 and published at ascopubs.org/journal/op on August 10, 2022; DOI <https://doi.org/10.1200/OP.22.00363>

Recommendation	Type	Evidence Quality	Strength
2.1. If a patient's HER2-positive advanced breast cancer has progressed during or after first-line HER2-targeted therapy (and the patient has not received trastuzumab deruxtecan), clinicians should recommend trastuzumab deruxtecan as a second-line treatment.	EB	M	S
Overall, in third-line there are a lack of head-to-head trials, therefore, there is insufficient evidence to recommend one regimen over another. The patient and clinician should discuss differences in treatment schedules, and routes, toxicities during the decision-making process. Options include the following:			
3.1. If a patient's HER2-positive advanced breast cancer has progressed during or after second-line or greater HER2-targeted treatment and the patient has already received pertuzumab and T-DM1, clinicians should recommend third-line or greater HER2-targeted therapy-based treatment.			
3.1.1. If a patient has not received T-DM1 in the second-line, should offer a T-DM1 regimen	EB	H	S
3.1.2. May offer tucatinib combined with trastuzumab and capecitabine	EB	M	S
3.1.3. May offer trastuzumab deruxtecan	EB	M	S
3.1.4. May offer neratinib plus capecitabine	EB	M	W
3.1.5. May offer lapatinib and trastuzumab	EB	M	W
3.1.6. May offer lapatinib and capecitabine	EB	M	W
3.1.7. May offer other combinations of chemotherapy and trastuzumab	EB	M	W
3.1.8. May offer margetuximab plus chemotherapy	EB	M	W
3.1.9. If a patient has not received pertuzumab, clinicians may offer pertuzumab	IC	Ins	W
3.2.0. May offer hormonal therapy (in patients with ER+ and/or PgR+ disease)	EB	M	W
3.2.1. May offer abemaciclib combined with trastuzumab and fulvestrant	EB	M	W

FIG 1. Summary of updated guideline recommendations. EB, evidence-based; ER+, estrogen receptor-positive; HER2, human epidermal growth factor receptor 2; H, high; IC, informal consensus; Ins, Insufficient; M, moderate; PgR+, progesterone receptor-positive; S, strong; T-DM1, trastuzumab emtansine; W, weak.

AFFILIATIONS

¹Seattle Cancer Care Alliance, Seattle, WA

²Hematology/Oncology Associates of Central New York, East Syracuse, NY

³American Society of Clinical Oncology, Alexandria, VA

CORRESPONDING AUTHOR

American Society of Clinical Oncology, 2318 Mill Rd, Suite 800, Alexandria, VA 22314; e-mail: guidelines@asco.org.

DISCLAIMER

This Q&A is derived from recommendations in Systemic Therapy for Advanced Human Epidermal Growth Factor Receptor 2-Positive Breast Cancer: ASCO Guideline Update. This document is based on an ASCO Guideline and is not intended to substitute for the independent professional judgment of the treating physician. Practice guidelines do not account for individual variation among patients. This Q&A does not

purport to suggest any particular course of medical treatment. Use of the guideline and this Q&A is voluntary. Please refer to the complete guideline to understand the full recommendations.

AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

Disclosures provided by the authors are available with this article at DOI <https://doi.org/10.1200/OP.22.00363>.

AUTHOR CONTRIBUTIONS

Conception and design: Jeffrey J. Kirshner, Sarah Temin

Administrative support: Sarah Temin

Collection and assembly of data: Sarah Temin

Data analysis and interpretation: All authors

Manuscript writing: All authors

Final approval of manuscript: All authors

Accountable for all aspects of the work: All authors

ACKNOWLEDGMENT

Systemic Therapy for Advanced Human Epidermal Growth Factor Receptor 2–Positive Breast Cancer: ASCO Guideline Update was developed and written by Sharon H. Giordano, MD, MPH; Maria Alice B. Franzoi, MD; Sarah Temin, MSPH; Carey K. Anders, MD; Sarat

Chandralapaty, MD, PhD; Jennie R. Crews, MD; Jeffrey J. Kirshner, MD; Ian E. Krop, MD, PhD; Nancy U. Lin, MD; Aki Morikawa, MD, PhD; Debra A. Patt, MD, MPH, MBA; Jane Perlmutter, PhD; Naren Ramakrishna, MD, PhD; and Nancy E. Davidson, MD.

REFERENCES

1. Giordano SH, Franzoi MAB, Temin S, et al: Systemic therapy for advanced human epidermal growth factor receptor 2–positive breast cancer: ASCO guideline update. *J Clin Oncol* 40:2612–2635, 2022
2. Ramakrishna N, Anders CK, Lin NU: Recommendations on disease management for patients with advanced human epidermal growth factor receptor 2–positive breast cancer and brain metastases: ASCO guideline update. *J Clin Oncol* 40:2636–2655, 2022
3. Cortes J, Kim SB, Chung WP, et al: Trastuzumab deruxtecan versus trastuzumab emtansine for breast cancer. *N Engl J Med* 386:1143–1154, 2022



ASCO® in Action

Your Source for Cancer Policy and Practice News

ASCO in Action provides the latest news and analysis on cancer policy and practice issues through a frequently updated newsfeed and biweekly newsletter. ASCO in Action provides key details for the cancer community on critical issues affecting the delivery of care, including federal funding for cancer research, the ongoing response to COVID-19, physician reimbursement, and more.

For more information, visit asco.org/ascoaction.

AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

Systemic Therapy for Advanced Human Epidermal Growth Factor Receptor 2–Positive Breast Cancer: ASCO Guideline Update Q and A

The following represents disclosure information provided by authors of this manuscript. All relationships are considered compensated unless otherwise noted. Relationships are self-held unless noted. I = Immediate Family Member, Inst = My Institution. Relationships may not relate to the subject matter of this manuscript. For more information about ASCO's conflict of interest policy, please refer to www.asco.org/rwc or ascopubs.org/op/authors/author-center.

Open Payments is a public database containing information reported by companies about payments made to US-licensed physicians ([Open Payments](#)).

No potential conflicts of interest were reported.